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A 6-9GHz DTC-based Fractional-N Sub-sampling PLL

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***Abstract*—In this paper, a digital-to-time converter (DTC)-based fractional-N sub-sampling phase locked loop (PLL) based on 28nm CMOS technology is presented. The PLL achieved an output frequency range of 6GHz to 9GHz with a power consumption of 37mW. Specifically, at an output frequency of 7.975GHz, the fractional spur with a 30% DTC gain error is -58.75dB. After gain calibration and threshold mismatch calibration, the fractional spur is reduced by 11dB to -69.8dB. Furthermore, the phase noise is -116dBc/Hz@1MHz, and the rms jitter is 113fs.**

Keywords—phase-locked-loop (PLL), fractional-N PLL, DTC gain calibration, threshold mismatch calibration, digital-to-time converter (DTC)

I. Introduction

In RF transceivers, a stable and low-noise local oscillator is crucial for ensuring the quality of the communication. Therefore, frequency synthesizers, which can generate multiple stable and accurate frequencies, have become the solution for producing high-performance multi-phase local oscillator in modern integrated circuits.

The mainstream architectures of PLL are charge pump PLL [1] and sub-sampling PLL [2]. The charge pump PLL has the advantages of small phase difference lock and stable structure, but it contributes large in-band phase noise that is difficult to suppress. On the other hand, the sub-sampling PLL has gained popularity due to its extremely low in-band noise.

The sub-sampling PLL directly samples the VCO or the high-speed signal after division with a low-frequency reference clock. Currently, most sub-sampling PLLs consist of a main phase lock loop and an auxiliary frequency locking loop. One notable feature of the sub-sampling PLL compared to the traditional charge-pump PLL is the removal of the divider in the loop.

The fractional-N sub-sampling PLL concept is primarily based on the DTC compensation-based fractional division scheme depicted in Fig 1. In this scheme, the divider in the feedback path is completely eliminated [3], reducing power consumption and decreasing in-band phase noise. At the same time, The time resolution of DTC can achieve femtosecond-level precision, allowing for finer adjustment of the fractional division ratio, achieving low in-band noise performance.

This paper proposes a DTC-based fractional-N sub-sampling PLL. A 10-bit constant slope CDAC-DTC circuit was designed to improve the DTC signal-to-noise ratio. The design

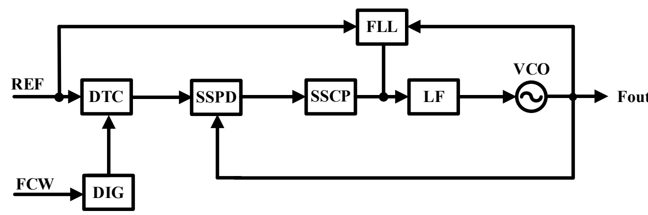


Fig 1. Implementation of fractional-N sub-sampling phase locked loop Based on DTC compensation.

employed segmented encoding and pseudo-random codes in the thermometer code to achieve this. The DTC has a delay time resolution of 201fs, a worst-case integral nonlinearity (INL) error of 1.3LSB, and a worst-case differential nonlinearity (DNL) error of 0.35LSB. To address DTC gain mismatch in fractional-N mode, a gain calibration algorithm based on least mean square (LMS) was designed. To overcome threshold mismatch of sub-sampling charge pump and comparator, a threshold calibration algorithm was implemented to ensure the accuracy of the gain calibration iterative function. When the output frequency is 7.975GHz, the spurious noise with a 30% gain error is reduced by 11dB to -69.8dB after gain calibration and threshold mismatch calibration. The phase noise is -116dBc/Hz@1MHz, and the rms jitter is 113fs.

II. PLL System and Circuits Design

Fig 2 shows the block diagram of the DTC-based fractional-N sub-sampling PLL. It includes a sub-sampling phase detector (SSPD) and a sub-sampling charge pump (SSCP), which converts the sampled voltage of the VCO output sampled by the low-frequency reference into current through the transconductance conversion of the SSCP. Then tune the 6-9GHz VCO for low power and low jitter requirements. The basic sub-sampling PLL cannot synthesize fractional-N frequencies, because it lacks any phase modulation mechanism in the loop.

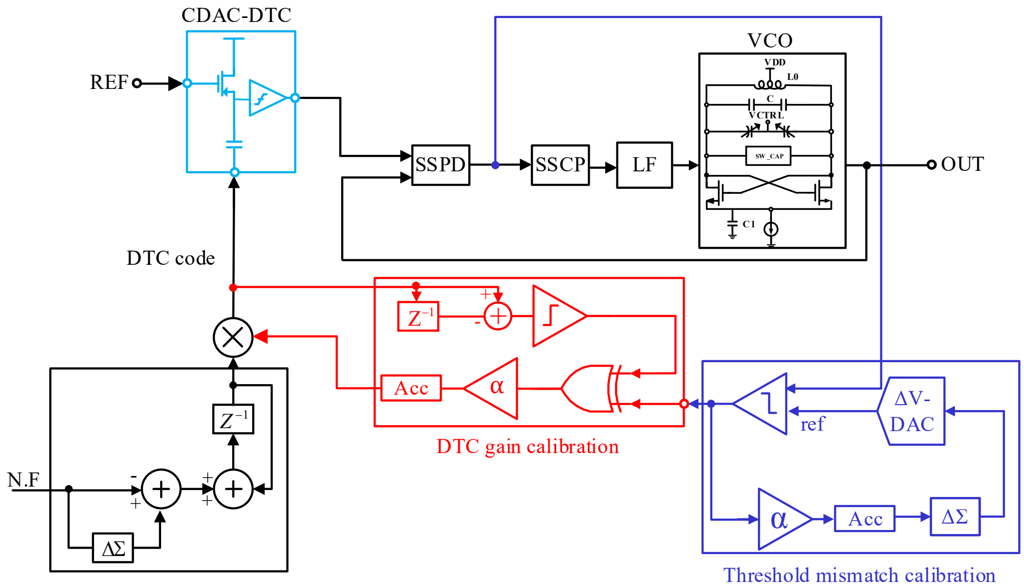


Fig. 2 Proposed DTC-based fractional-N sub-sampling PLL.

So, DTC is introduced in the loop to achieve phase modulation. The DTC will be introduced in the following sections of this chapter.

The self-calibration algorithm includes DTC gain calibration and threshold mismatch calibration. In order to calibrate the gain error of DTC in real-time, a gain calibration algorithm based on LMS was designed. Due to the influence of mismatch in analog circuits on the error term in the LMS algorithm, a threshold mismatch calibration algorithm has been designed for the error term. The digital algorithm will be introduced in the following sections of this chapter.

A. CDAC-DTC

The DTC circuit is controlled by a digital control word (DCW) and an input reference clock. The DCW adjusts the DTC delay step through a decoding circuit and a switch array, so that the DTC has different delays when different input DCW are used [4].

The overall structure of DTC is shown in Fig 4. In this design, a capacitive DAC, mainly based on its high-speed and high-precision characteristics is chosen. The input of CADC is a multi-bit DCW, and the output is used to drive the threshold comparator switch. The traditional binary weighted CDAC consists of capacitor units arranged according to binary weights(C_U, C_U, $2C_U$, 2^2C_U,..., $2^{n-1}C_U$), and the sum of the capacitor units constitutes the total capacitance value of the capacitor array.

Fig 3 shows the schematic of CDAC, assuming there are N capacitor units, where M capacitors are connected to the reference and the remaining (N-M) capacitors are grounded. Assuming that the parasitic capacitance value connected to the output node is not considered, the output generated by the series voltage division principle of capacitors can be calculated using the following formula:

$$V_{out} = V_{ref}\frac{C_M}{C_{total}} = V_{ref}\frac{\sum_{i=1}^{m}\alpha_i C_U}{2^n C_U} \tag{1}$$

Where α is the weight of the digital control bit.

For the coding method of CDAC, this article chose a scheme with a thermometer structure accounting for 40%. For a 10 bit CDAC, a segmented structure of "6+4" was employed. The low bit [5:0] uses binary code, while the high bit [9:6] uses a thermometer code converted to 15 bits through encoding.

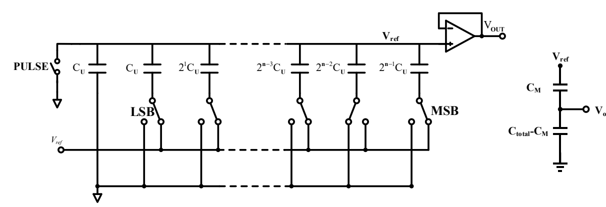


Fig. 3. schematic of CDAC.

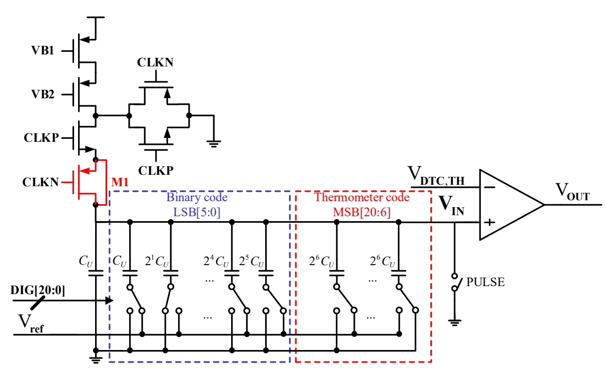


Fig. 4. Proposed DTC structure.

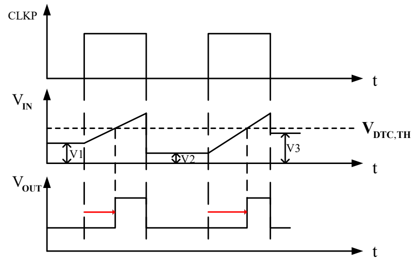


Fig. 5. DTC timing diagram.

During CDAC operation, the resistive load connected to the output terminal extracts charge from the other end of the capacitor, thereby affecting the performance and accuracy of the CDAC. To ensure that CDAC can accurately drive resistive loads, an amplifier with a large input impedance is introduced at the output end to serve as a threshold comparator. This can effectively isolate the impact of the load on the internal charge transfer process of the DAC. Using a common source and common gate current source to charge the initial voltage platform generated by CDAC, due to its high impedance characteristics, it maintains high-precision current values at different initial voltages of CDAC, generating a constant slope voltage ramp.

Fig 5 illustrates the timing diagram of DTC circuit operation. V1, V2, and V3 represent initial voltages of CDAC corresponding to different DTC digital control words. During the switching of the charging time, the clock CLK can be coupled to the output due to the clock feedthrough effect, resulting in charge injection into the channel and introducing offset in the current output. To counteract the channel charge injected by the switching transistors, a PMOS transistor M1 with its source and drain shorted is introduced below the switch terminal. Its gate is controlled by CLKN.

B. Self-calibration algorithm

The self-calibration algorithm includes DTC gain calibration and threshold calibration. By applying this algorithm, real-time tracking of changes in DTC gain can be achieved, and the impact of mismatch of SSCP and comparator can be reduced, significantly improving the robustness and performance stability of the system.

The basic form of DTC gain calibration is based on an LMS-regression described in [5] and [6].The specific block form for implementing the automatic gain calibration function of DTC using LMS algorithm in this article is shown in Fig 6.

The algorithm scales the DTC input by the calibration coefficient before being input to the sub-sampling PLL for sampling, with a certain delay. Due to DTC gain error, the actual delay time of DTC may be shorter than expected, causing the SSPD to provide an advanced sampling value V_{SAMP} as feedback. By detecting the sampling voltage of the SSPD, whether the gain of the current DTC is appropriate can be determined. In fact, a quantizer can be used to analyze the polarity of V_{SAMP}, which not only improves the reliability of the system but also effectively reduces hardware costs.

As shown in Fig 7, with a 10% gain error in the DTC control word, the sub-sampling PLL locks after 3μs, and the gain

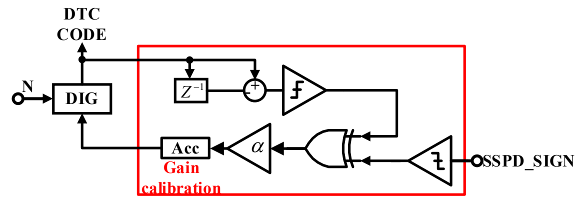


Fig. 6. DTC Gain calibration.

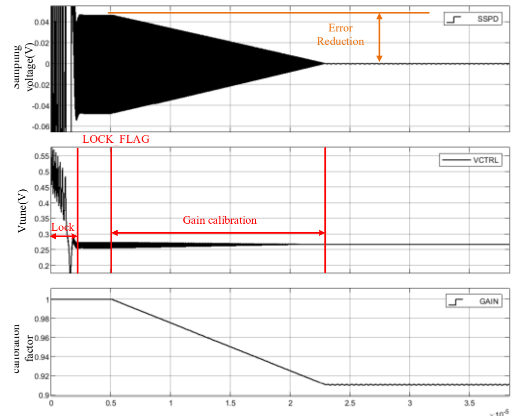


Fig. 7. Gain calibration process.

calibration algorithm was controlled to start working at 5μs. After 18μs, the gain calibration algorithm converged and the calibration coefficient converged to 0.91. However, due to limited accuracy, the calibration coefficient still exhibits small fluctuations. After the convergence of the gain calibration algorithm, the sampling error voltage fluctuation range is reduced from 50mV to around 1mV, proving the significance of the gain calibration algorithm for DTC.

As shown in Fig 8, the sub-sampling charge pump and comparator encounter voltage mismatch, potentially causing the error symbol extracted by the locked SSPD to deviate from absolute 0 or 1 [7]. Therefore, a threshold mismatch calibration algorithm was proposed, where the output of the comparator was multiplied by a scaling factor α and accumulated the threshold mismatch coefficient γ.

As shown in Fig 9, the processed mismatch coefficient γ is quantized by the ΔΣ modulator. The quantization result of the first-order ΔΣ modulator is the average value of the mismatch coefficient. The quantized output is used to modulate the ΔV-DAC, causing the average output of the ΔV-DAC to shift, thereby obtaining the calibrated comparator threshold voltage.

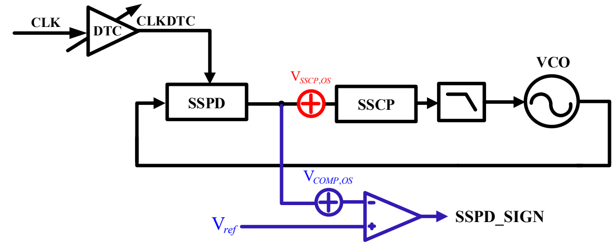


Fig. 8. Voltage mismatch phenomenon of the comparator and sub-sampling phase detector.

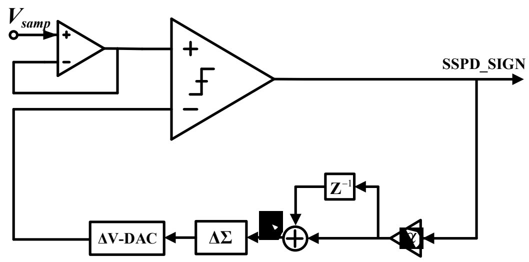


Fig. 9 Structure diagram of comparator threshold mismatch calibration algorithm.

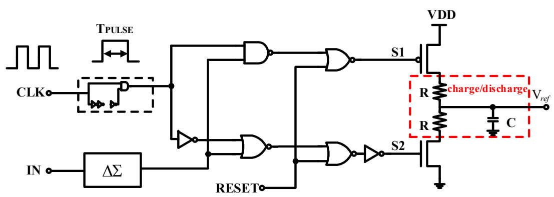


Fig. 10 ΔV-DAC circuit structure diagram.

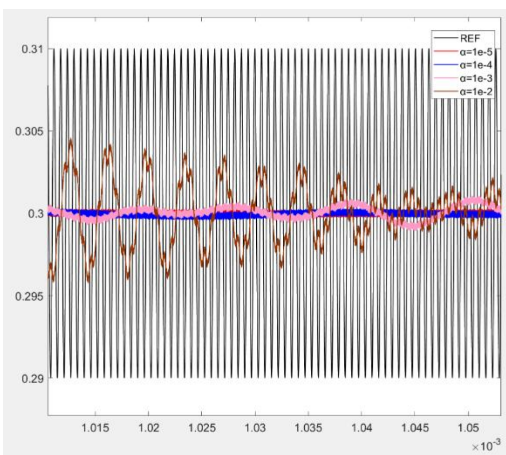


Fig. 11 Calibration of AC mismatch with reference voltage.

The function of ΔV-DAC in this module is to respond to the input and adjust the output voltage ΔV based on the polarity and size of the input. The circuit structure of ΔV-DAC is shown in Fig 10. The pulse generator is used to generate a pulse signal with adjustable duty cycle for a single clock cycle. During the high-level duration, the charge pump changes the input of the RC integrator based on the digital input, and the RC integrator continuously adjusts the reference voltage during the pulse duration. The calibration process under different scaling factors is shown in Fig 11.

III. Simulation Result

The layout design of the DTC based fractional-N sub-sampling PLL is shown in the Fig 12,with a total area of 450μm * 840μm. The power supply of the PLL is 1V. And the VCO power supply is independent of other circuits.

As shown in Fig 13, the DTC delay range is 0-206ps, with a unit delay time of 201fs. The worst INL is 1.3LSB, and the worst DNL is 0.35LSB.

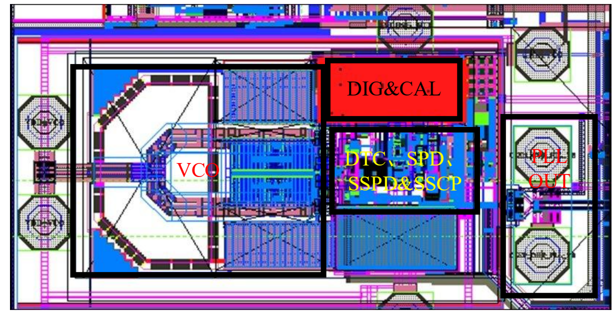
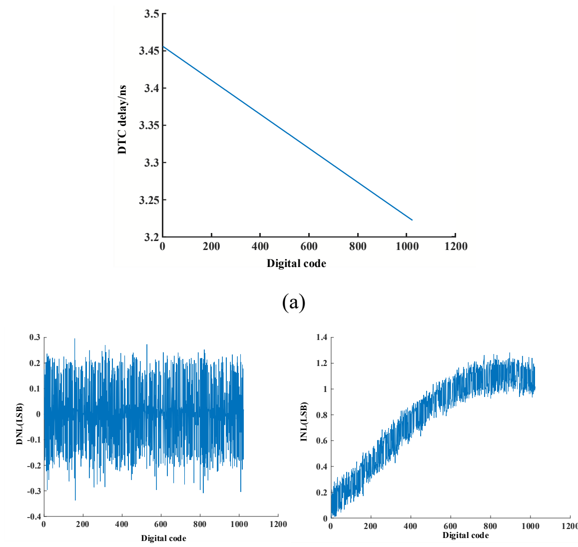


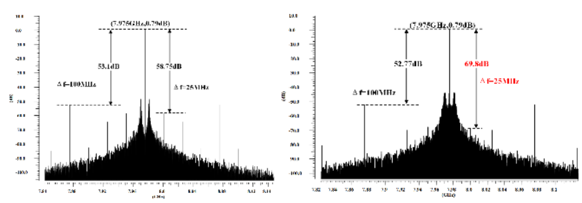
Fig. 12 Layout of the DTC based PLL.



(a)

(b) (c)

Fig. 13 (a) Simulation results of DTC circuit delay time.(b) DTC circuit DNL simulation results.(c) DTC circuit INL simulation results.



(a) (b)

Fig. 14 (a) Simulation output spectrum after 30% gain error.(b) Simulated output spectrum after gain calibration.

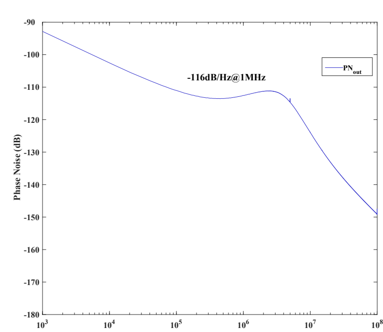


Fig. 15 Simulated output phase noise

TABLE I. PERFORMANCE SUMMARY AND COMPARION

	This work	[7]	[8]	[9]	[10]
Reference(MHz)	100	153.6	125	100	250
Output freq(GHz)	6-9	5-7	8.75-10.25	3.3-4.5	11.1-14.9
Supply voltage(V)	1	1	N/A	N/A	N/A
Technology	28 nm	14 nm	28 nm	40 nm	28 nm
rms jitter(fs)	113	80	66.7	292	41.3
Fractional spur(dBc)	-69	-72	-63.8	-54	-66.9
Power(mW)	37	14.2	16.7	2.4	34
FOM(dB)	-243.3	-250.4	-251.3	-246.9	-252.4

$\mathrm{FOM} = 20\log_{10}(\sigma t / 1s) + 10\log_{10}(\mathrm{Power} / 1mW)$

When the reference frequency is 100MHz and the division ratio is 79.75, the output frequency is 7.975GHz. FFT analysis was performed on the time-domain waveform before and after gain calibration, and the post simulation output spectrum results before and after calibration are shown in Fig 14. Before calibration, the fractional spur at the 25MHz frequency offset was -58.75dB in Fig 14(a). After gain calibration, the fractional spur decreased by 11.05dB to -69.8dB in Fig 14(b).

Simulate the phase noise of DTC, SSPD, SSCP, and VCO in the sub-sampling PLL and fit them in MATLAB to obtain the equivalent phase noise curve to the output. As shown in Fig 15, when the output frequency is 7.975GHz, it can be seen that the phase noise at the 1MHz frequency offset is -116dBc/Hz. By integrating the phase noise within the 1k-100MHz range, the rms jitter can be obtained as 113fs. Table I shows the performance summary and comparison to other fractional-N PLLs. This PLL achieves 113fs rms jitter, and a power jitter FOM of -243.3dB, which is comparable to the performance of them. Compared with [8], [9], and [10], the fractional spur is -69dBc, better than them.

IV. CONCLUSION

This paper proposes a DTC-based fractional-N PLL, where a DTC architecture based on constant slope is selected, and the CDAC structure is designed accordingly. A binary code to thermometer code segmentation ratio for the encoding in CADC is designed, and use pseudo-random code encoding for thermometers to increase pseudo-random interference. The DTC delay range is 0-206ps, with a unit delay time of 201fs. The worst INL is 1.3LSB, and the worst DNL is 0.35LSB. When the output of the PLL is 7.975GHz, DTC gain calibration is performed to reduce the fractional spur from 58.75dB to -69.8dB when there is a 30% gain error. The noise fitting of the sub-sampling PLL in MATLAB resulted in a phase noise of -116dBc/Hz @1MHz for the designed PLL when the output frequency is 7.975GHz. It can achieve 113fs of rms jitter.

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